

Integrative Mendelian randomization for detecting exposure-by-group interactions using group-specific and combined summary statistics

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Introduction

This vignette provides an introduction to the `int2MR` R package. The `int2MR` package implements the `int2MR` method for detecting group-specific exposure-outcome effects and exposure-group interaction effects using summary statistics.

Before installing `int2MR`, ensure that the `devtools` package is installed. `int2MR` also requires `rstan` for Bayesian modeling. To install and load the development version of `int2MR`, run:

```
# Load devtools package
suppressMessages(library(devtools))

# Install int2MR from GitHub
install_github("kxu-stat/int2MR")

# Load int2MR
library(int2MR)

# Load rstan
suppressMessages(library(rstan))
```

Input Data Format

The `int2MR` package supports two types of input:

- **Two-sample data:** For integrating two sets of IV-to-outcome GWAS summary statistics.
- **Three-sample data:** For Integrating three sets of IV-to-outcome GWAS summary statistics.

Two-sample input data must be a list with the following components: `p` is the number of IVs; `hat_gamma` and `hat_s_gamma_sq` are the estimated IV-to-exposure effects and their variances (squared standard errors); `hat_Gamma1` and `hat_s1_sq` are the estimated IV-to-outcome effects for the first IV-to-outcome GWAS summary statistic and their variances; `hat_Gamma2` and `hat_s2_sq` are the estimated IV-to-outcome effects for the second IV-to-outcome GWAS summary statistic and their variances; `rho1` is the proportion of the comparison group in the sample of the first GWAS summary statistic; and `rho2` is the proportion of the comparison group in the sample of the second GWAS summary statistic.

```
# Load the example two-sample data
data(example_2sample_data)
names(example_2sample_data)
```

Three-sample input data additionally contains `hat_Gamma3`, `hat_s3_sq`, and `rho3` for the third GWAS summary statistic.

```
# Load the example three-sample data
data(example_3sample_data)
names(example_3sample_data)
```

To detect the exposure-group interaction effect, the proportion of the comparison group in the samples (`rho`) must vary among the provided GWAS summary statistics.

Running Examples

This section demonstrates how to run `int2MR` using simulated data. Two examples are provided: one for two-sample data and one for three-sample data.

Example 1: Two-Sample Data

In this example, we perform an analysis on simulated two-sample data. The parameters include:

- **data_list_2sample:** Input data for two-sample analysis. Note that the `data_list_3sample` argument is left null
- **model_type:** “2sample”
- **estimation:** “sampling” for MCMC posterior mean effect estimates, or “optimizing” for optimization-based MAP effect estimates
- **Prior distributions:** Inverse gamma priors with a shape and scale of 0.02
- **MCMC Settings:** 2 chains, 10,000 iterations with a warm-up period of 2,500 iterations, and an `adapt_delta` of 0.95.

The inverse-gamma hyperparameters control the prior scale of uncorrelated pleiotropic-effect variance components. As a rule of thumb, start with `prior_inv_gamma_shape = prior_inv_gamma_scale = 0.02` for weak or sparse uncorrelated pleiotropic effects, try 0.1 for moderate effects, and check stronger effects with a small sensitivity grid such as `c(0.02, 0.05, 0.1, 0.2, 0.5)`. Report whether `est_beta` and `est_beta_int` are stable across the grid rather than relying on a single setting.

```
result_2sample <- int2MR(data_list_2sample = example_2sample_data,
  model_type = "2sample",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02,
  chains = 2, iter = 10000, warmup = 2500,
  adapt_delta = 0.95)

# Display the results for the two-sample analysis
result_2sample$result_2sample
```

`est_beta` is the estimated group-specific exposure-outcome effect for the reference group, and `se_beta` and `pval_beta` are its standard error and p-value, respectively. `est_beta_int` is the estimated exposure-group interaction effect, and `se_beta_int` and `pval_beta_int` are its standard error and p-value, respectively. `total_effect` is the estimated group-specific exposure-outcome effect for the comparison group and `pval_total` is its respective p-value.

To estimate the effects without MCMC sampling, use `estimation = "optimizing"`. The `estimation` argument controls how `est_beta` and `est_beta_int` are estimated: sampling mode uses MCMC posterior means, while optimizing mode uses the MAP estimates. In both modes, `se_beta`, `se_beta_int`, and `se_total` are computed from the inverse negative Hessian.

```
result_2sample_opt <- int2MR(data_list_2sample = example_2sample_data,
  model_type = "2sample",
  estimation = "optimizing",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02)

result_2sample_opt$result_2sample
```

Example 2: Three-Sample Data

In this example, we perform an analysis on simulated three-sample data. The configuration is similar to the two-sample analysis, except the `data_list_3sample` argument is used instead of `data_list_2sample` and `model_type` is set to "3sample".

```
result_3sample <- int2MR(data_list_3sample = example_3sample_data,
  model_type = "3sample",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02,
  chains = 2, iter = 10000, warmup = 2500,
  adapt_delta = 0.95)

# Display the results for the three-sample analysis
result_3sample$result_3sample
```

The three-sample model can also be run with `estimation = "optimizing"` to use the same optimization-based estimation path.

Choosing Between Sampling and Optimization

For routine reporting, MCMC sampling is the default because it estimates `est_beta` and `est_beta_int` using posterior means. Optimization is useful for fast screening, sensitivity analyses over many hyperparameter settings, or large-scale scans where running MCMC for every analysis is computationally expensive. In both modes, `se_beta`, `se_beta_int`, and `se_total` are computed from the inverse negative Hessian.

```
result_sampling <- int2MR(data_list_2sample = example_2sample_data,
  model_type = "2sample",
  estimation = "sampling",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02,
  chains = 2, iter = 10000, warmup = 2500,
  adapt_delta = 0.95)
```

```

result_optimizing <- int2MR(data_list_2sample = example_2sample_data,
  model_type = "2sample",
  estimation = "optimizing",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02)

rbind(
  sampling = result_sampling$result_2sample,
  optimizing = result_optimizing$result_2sample
)

```

Hyperparameter Sensitivity Grid

When the expected strength of uncorrelated pleiotropic effects is uncertain, repeat the analysis over a small grid of inverse-gamma hyperparameters. Stable `est_beta` and `est_beta_int` estimates across the grid provide more reassuring evidence than results that appear only under one prior setting.

```

prior_grid <- c(0.02, 0.05, 0.1, 0.2, 0.5)

sensitivity_results <- do.call(rbind, lapply(prior_grid, function(prior_value) {
  fit <- int2MR(data_list_2sample = example_2sample_data,
    model_type = "2sample",
    estimation = "optimizing",
    prior_inv_gamma_shape = prior_value,
    prior_inv_gamma_scale = prior_value)

  cbind(prior_value = prior_value, fit$result_2sample)
}))

sensitivity_results[, c("prior_value", "est_beta", "se_beta",
  "est_beta_int", "se_beta_int",
  "total_effect", "se_total")]

```

For stronger suspected uncorrelated pleiotropic effects, users can focus more attention on the upper part of the grid, such as 0.1, 0.2, and 0.5. If the sign, magnitude, or significance of `est_beta_int` changes materially across the grid, report the sensitivity rather than presenting one prior choice as definitive.

Running Two-Sample and Three-Sample Analyses Together

If both data objects are available, `model_type = "both"` runs the two-sample and three-sample models in one call. This is useful when comparing whether the inferred exposure effect and exposure-by-group interaction are consistent across input designs.

```

result_both <- int2MR(data_list_2sample = example_2sample_data,
  data_list_3sample = example_3sample_data,
  model_type = "both",
  estimation = "optimizing",
  prior_inv_gamma_shape = 0.02,
  prior_inv_gamma_scale = 0.02)

```

```
result_both$result_2sample  
result_both$result_3sample
```